TWS Bluetooth Audio Manual Testing Lab

30-Day Self-Learning Training Program

Manual Testing • Bluetooth • Audio QA • Test Design • Defect Management • Reporting

Practical Training Manual

A structured, vendor-neutral self-learning program for developing practical skills in TWS/Bluetooth audio manual testing using commercially available hardware.

Recommended Test Setup

Windows laptop • Android smartphone • TWS earbuds (DUT) • Excel/Sheets • VS Code • Python • Git/GitHub

Project Repository

TWS-Bluetooth-Audio-Manual-Testing-Lab

1. Program Overview

This manual is designed to be followed sequentially over 30 days. Each day combines study, hands-on testing, documentation, and a GitHub deliverable. The goal is to simulate a professional QA workflow from research through final reporting.

Core workflow

Research → Requirements → Test Strategy → Test Plan → Test Case Development → Test Execution → Bug Reporting → Bug Validation → Regression → Code Review → Failure Analysis → Test Summary Report

Learning outcomes

• Understand TWS architecture and Bluetooth audio concepts.

• Develop structured manual test cases and scenarios.

• Perform functional, exploratory, negative, connectivity and regression testing.

• Write reproducible and professional defect reports.

• Validate fixes and distinguish retesting from regression testing.

• Create test strategy, test plan and requirements traceability documentation.

• Perform basic test-related Python code review and log/failure analysis.

• Prepare a professional Test Summary Report and QA portfolio.

2. Ground Rules

• Use actual observations; do not invent expected or actual results.

• Do not call an observation a defect until the expected behavior/requirement is established.

• Record environment details for every significant failure.

• Use reproducible steps and timestamps where useful.

• Separate retest from regression testing.

• Keep the repository vendor-neutral and avoid implying manufacturer affiliation.

• Commit meaningful work to GitHub regularly.

3. Recommended Repository Structure

```
TWS-Bluetooth-Audio-Manual-Testing-Lab/
■■■ README.md
■■■ 01-Research/
■■■ 02-Requirements/
■■■ 03-Test-Strategy/
■■■ 04-Test-Plan/
■■■ 05-Test-Cases/
■■■ 06-Test-Execution/
■■■ 07-Bugs/
■■■ 08-Bug-Validation/
■■■ 09-Code-Review/
■■■ 10-Failure-Analysis/
■■■ 11-RTM/
■■■ 12-Test-Summary/
```

4. Daily Working Method

1. Study the day's concepts for approximately 45–60 minutes.
2. Perform hands-on work for approximately 60–120 minutes.
3. Record observations immediately.
4. Create/update the specified GitHub artifact.
5. Review your work for clarity and evidence.
6. Commit with a descriptive Day-N commit message.
7. Mark the day's checklist complete.

Day 01 — Project Foundation & QA Orientation

Objective: Understand the objective, DUT, test environment and basic QA vocabulary.

Study Topics

• QA vs QC vs testing

• verification vs validation

• DUT, test environment, defect, expected vs actual result

• basic STLC and defect lifecycle

Hands-on Activities

1. Create the GitHub repository.

2. Create 01-Research and 02-Requirements.

3. Document Windows laptop, Android phone and TWS DUT.

4. Perform initial Bluetooth discovery and pairing observation.

5. Record unknowns without labeling them defects.

GitHub Deliverables

• 01-Research/README.md

• 01-Research/Test-Environment.md

• 01-Research/DUT-Behavior-Notes.md

• 02-Requirements/README.md

Suggested Commit: Day 1: Complete project foundation and initial research

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 02 — TWS Architecture & Bluetooth Fundamentals

Objective: Understand the main components and Bluetooth profiles relevant to TWS testing.

Study Topics

• TWS architecture

• A2DP audio streaming

• AVRCP media control

• HFP voice/calls

• GAP and GATT basics

• pairing vs connection

Hands-on Activities

1. Draw a simplified TWS architecture.

2. Study the audio, control and call paths.

3. Perform discovery, pairing, connection and disconnect/reconnect observations.

4. Record actual device behavior.

GitHub Deliverables

• 01-Research/Bluetooth-Fundamentals.md

• 01-Research/TWS-Architecture.md

• Update DUT-Behavior-Notes.md

Suggested Commit: Day 2: Add TWS architecture and Bluetooth fundamentals

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 03 — Bluetooth Connection States & Scenarios

Objective: Build a state-based understanding of discovery, pairing, connection and recovery.

Study Topics

• discovery

• pairing

• connected

• disconnected

• out of range

• reconnection

• re-pairing

Hands-on Activities

1. Create a Bluetooth state diagram.

2. Explore first-time pairing, previously paired connection, unpair/re-pair and Bluetooth OFF/ON.

3. Record transition behavior and timing where useful.

GitHub Deliverables

• 01-Research/Bluetooth-Connection-States.md

• Update DUT-Behavior-Notes.md

Suggested Commit: Day 3: Document Bluetooth connection states

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 04 — Audio Fundamentals

Objective: Understand audio concepts that affect manual validation.

Study Topics

• sample rate

• bit depth

• PCM

• codec basics

• bitrate

• frequency

• latency

• distortion and noise

Hands-on Activities

1. Use known audio files.

2. Compare speech, music, silence, low-frequency and high-frequency content.

3. Use an audio editor if available to inspect files.

4. Record perceived issues separately from measurable facts.

GitHub Deliverables

• 01-Research/Audio-Fundamentals.md

• 01-Research/Audio-Test-Materials.md

Suggested Commit: Day 4: Add audio fundamentals and test materials

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 05 — Bluetooth Discovery & Pairing Test Design

Objective: Learn to convert behavior into test scenarios.

Study Topics

• test scenario vs test case

• preconditions

• steps

• expected result

• priority

Hands-on Activities

1. Draft 15–20 pairing/discovery scenarios.

2. Include first pairing, re-pairing, wrong device, Bluetooth disabled, multiple nearby devices and failed/aborted pairing.

GitHub Deliverables

• 05-Test-Cases/Pairing-Test-Cases.xlsx

Suggested Commit: Day 5: Create Bluetooth pairing test cases

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 06 — Audio Playback Test Design

Objective: Design structured audio playback coverage.

Study Topics

• play/pause/resume

• volume

• track control

• left/right audio

• interruption and recovery

Hands-on Activities

1. Create 15–20 audio cases.

2. Include normal, boundary and negative scenarios.

3. Define precise expected results.

GitHub Deliverables

• 05-Test-Cases/Audio-Test-Cases.xlsx

Suggested Commit: Day 6: Create audio playback test cases

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 07 — Week 1 Mini Test Cycle

Objective: Perform the first controlled manual test cycle.

Study Topics

• smoke testing

• test execution discipline

• evidence collection

Hands-on Activities

1. Execute a small suite across Bluetooth and audio.

2. Record Pass/Fail/Blocked/Not Run.

3. Capture evidence for failures or unexpected behavior.

4. Review case quality.

GitHub Deliverables

• 06-Test-Execution/Week-01-Execution.xlsx

• 06-Test-Execution/Week-01-Report.md

Suggested Commit: Day 7: Complete first manual test cycle

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 08 — Call & Microphone Test Design

Objective: Create professional call and microphone scenarios.

Study Topics

• incoming/outgoing calls

• answer/reject/end

• microphone

• voice clarity

• music-call transitions

Hands-on Activities

1. Create 15–20 call cases.

2. Test music→call and call→music transitions.

3. Record host and DUT behavior.

GitHub Deliverables

• 05-Test-Cases/Call-Microphone-Test-Cases.xlsx

Suggested Commit: Day 8: Create call and microphone test cases

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 09 — Connectivity & Reconnection Testing

Objective: Explore recovery from connectivity changes.

Study Topics

• Bluetooth OFF/ON

• phone reboot

• earbud restart

• out-of-range

• return-to-range

• multiple devices

Hands-on Activities

1. Create 20 connectivity/reconnection cases.

2. Execute the most important scenarios.

3. Record whether recovery is automatic or manual.

GitHub Deliverables

• 05-Test-Cases/Connectivity-Test-Cases.xlsx

• 06-Test-Execution/Connectivity-Execution.xlsx

Suggested Commit: Day 9: Add connectivity and reconnection coverage

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 10 — Negative Testing

Objective: Learn to deliberately challenge normal behavior.

Study Topics

• negative testing

• error handling

• abnormal sequences

• recovery

Hands-on Activities

1. Create 20 negative cases.

2. Interrupt playback, calls and Bluetooth connections.

3. Test low battery and repeated connect/disconnect where practical.

4. Document observations without premature defect labels.

GitHub Deliverables

• 05-Test-Cases/Negative-Test-Cases.xlsx

• 06-Test-Execution/Negative-Execution.xlsx

Suggested Commit: Day 10: Complete negative testing exercise

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 11 — Test Case Design Review

Objective: Review the growing test suite for quality and coverage.

Study Topics

• test case quality

• duplicates

• ambiguity

• boundary coverage

• traceability

Hands-on Activities

1. Review all cases.

2. Remove duplicates.

3. Make steps reproducible.

4. Ensure expected results are observable and specific.

5. Target 60–80 cumulative cases.

GitHub Deliverables

• 05-Test-Cases/TWS-Test-Cases.xlsx

• 05-Test-Cases/Test-Case-Review.md

Suggested Commit: Day 11: Review and consolidate test cases

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 12 — Exploratory Testing

Objective: Develop tester intuition beyond predefined scripts.

Study Topics

• charters

• session-based exploration

• heuristics

• observations

Hands-on Activities

1. Run 2 exploratory sessions of 30–45 minutes.

2. Try unusual sequences and combinations.

3. Record discoveries, questions and possible defects.

GitHub Deliverables

• 06-Test-Execution/Exploratory-Test-Charter.md

• 06-Test-Execution/Exploratory-Session-01.md

Suggested Commit: Day 12: Perform exploratory testing

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 13 — Test Data & Evidence

Objective: Standardize evidence collection.

Study Topics

• test data

• screenshots

• timestamps

• environment metadata

• repro evidence

Hands-on Activities

1. Create a naming convention for evidence.

2. Prepare standard test data: audio files, call scenarios and device states.

3. Document how evidence is stored.

GitHub Deliverables

• 06-Test-Execution/Test-Data-and-Evidence.md

Suggested Commit: Day 13: Standardize test data and evidence

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 14 — Week 2 Test Suite Baseline

Objective: Freeze a clean baseline of the initial test suite.

Study Topics

• baseline

• coverage

• review sign-off mindset

Hands-on Activities

1. Consolidate all cases into one master sheet.

2. Target 80–100 cases.

3. Categorize by Bluetooth, audio, calls, connectivity and negative testing.

4. Prepare for execution.

GitHub Deliverables

• 05-Test-Cases/TWS-Test-Cases.xlsx

• 05-Test-Cases/Coverage-Matrix.md

Suggested Commit: Day 14: Establish test suite baseline

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 15 — Formal Test Execution

Objective: Execute the baseline suite systematically.

Study Topics

• Pass

• Fail

• Blocked

• Not Run

• actual result

Hands-on Activities

1. Execute a meaningful subset or full suite depending on time.

2. Record actual results and evidence.

3. Do not change expected results to match observations.

GitHub Deliverables

• 06-Test-Execution/Test-Execution.xlsx

Suggested Commit: Day 15: Start formal test execution

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 16 — Professional Bug Reporting

Objective: Learn reproducible defect documentation.

Study Topics

• bug summary

• environment

• preconditions

• steps

• expected vs actual

• severity

• priority

• reproducibility

Hands-on Activities

1. Convert confirmed failures into defect reports.

2. Write concise summaries.

3. Include exact reproduction steps and evidence.

GitHub Deliverables

• 07-Bugs/Bug-Reports.xlsx

• 07-Bugs/Bug-Reporting-Guidelines.md

Suggested Commit: Day 16: Add professional defect reporting

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 17 — Bug Hunting Session

Objective: Perform focused exploratory defect discovery.

Study Topics

• bug hunting

• reproducibility

• intermittent failures

• observation vs defect

Hands-on Activities

1. Run a 2–3 hour focused session.

2. Try high-risk transitions.

3. Document 5–10 candidate observations; classify only after validation.

GitHub Deliverables

• 07-Bugs/Bug-Hunting-Session.md

• 07-Bugs/Bug-Candidates.xlsx

Suggested Commit: Day 17: Perform focused bug-hunting session

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 18 — Severity & Priority

Objective: Learn to classify defects consistently.

Study Topics

• severity: blocker/critical/major/minor/trivial

• priority: P0/P1/P2/P3

Hands-on Activities

1. Review each defect.

2. Justify severity and priority separately.

3. Identify user and release impact.

GitHub Deliverables

• 07-Bugs/Severity-Priority-Matrix.md

Suggested Commit: Day 18: Classify defect severity and priority

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 19 — Bug Reproduction & Failure Analysis

Objective: Determine whether reported issues are reproducible.

Study Topics

• reproduction rate

• frequency

• environment dependency

• failure timeline

Hands-on Activities

1. Repeat candidate bugs multiple times.

2. Record pass/fail frequency.

3. Capture exact conditions preceding failure.

GitHub Deliverables

• 07-Bugs/Bug-Reproduction-Log.xlsx

• 10-Failure-Analysis/Initial-Failure-Analysis.md

Suggested Commit: Day 19: Validate defect reproducibility

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 20 — Retest & Bug Validation

Objective: Learn how to validate a simulated fix.

Study Topics

• retest

• fixed/not fixed

• reopened

• validation evidence

Hands-on Activities

1. For each selected defect, define a retest case.

2. Simulate or use a changed condition where appropriate.

3. Record validation outcome.

GitHub Deliverables

• 08-Bug-Validation/Retest-Report.xlsx

Suggested Commit: Day 20: Perform bug validation and retest

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 21 — Regression Testing

Objective: Understand impact-based regression.

Study Topics

• regression scope

• affected area

• related functionality

• smoke vs regression

Hands-on Activities

1. Select regression cases around each fixed defect.

2. Execute them.

3. Document whether the fix caused side effects.

GitHub Deliverables

• 08-Bug-Validation/Regression-Report.xlsx

Suggested Commit: Day 21: Complete regression testing cycle

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 22 — Test Strategy

Objective: Design the overall QA approach before a release.

Study Topics

• scope

• test types

• risks

• dependencies

• entry/exit criteria

Hands-on Activities

1. Write a strategy for the TWS product domain.

2. Define functional, negative, exploratory, connectivity and regression approaches.

3. Identify risks and mitigations.

GitHub Deliverables

- 03-Test-Strategy/Test-Strategy.md

Suggested Commit: Day 22: Create test strategy

Completion Checklist

- Study completed
- Hands-on activity completed
- Actual observations recorded
- GitHub documentation updated
- Work reviewed for accuracy and clarity
- Commit created

Day 23 — Test Plan

Objective: Turn the strategy into an executable plan.

Study Topics

- schedule
- resources
- environment
- deliverables
- roles
- entry/exit criteria

Hands-on Activities

1. Create a test plan covering setup, execution, defect handling and reporting.
2. Define what constitutes completion.

GitHub Deliverables

- 04-Test-Plan/Test-Plan.md

Suggested Commit: Day 23: Create test plan

Completion Checklist

- Study completed
- Hands-on activity completed
- Actual observations recorded
- GitHub documentation updated
- Work reviewed for accuracy and clarity
- Commit created

Day 24 — Requirements Analysis & RTM

Objective: Connect requirements to tests and defects.

Study Topics

• requirements

• test coverage

• RTM

• bidirectional traceability

Hands-on Activities

1. Create 15–25 realistic product requirements.

2. Map each requirement to one or more test cases.

3. Link defects where applicable.

GitHub Deliverables

• 02-Requirements/Requirements.md

• 11-RTM/RTM.xlsx

Suggested Commit: Day 24: Build requirements and traceability

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 25 — Python Code Review

Objective: Develop basic code-review skills for test utilities.

Study Topics

• logic

• exceptions

• input validation

• naming

• maintainability

• boundary conditions

Hands-on Activities

1. Create or review small Python utilities for parsing test results, calculating pass rates or generating CSV summaries.

2. Identify defects and improvement opportunities.

GitHub Deliverables

• 09-Code-Review/Python-Code/

• 09-Code-Review/Code-Review.md

Suggested Commit: Day 25: Complete Python code review exercise

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 26 — Logs & Failure Analysis

Objective: Use logs and timelines to investigate failures.

Study Topics

• timestamps

• events

• timeouts

• retries

• correlation

Hands-on Activities

1. Collect available Windows/app/device information.

2. Build a failure timeline.

3. Separate evidence from hypotheses.

4. Write possible next investigation steps.

GitHub Deliverables

• 10-Failure-Analysis/Failure-Analysis.md

• 10-Failure-Analysis/Failure-Timeline.xlsx

Suggested Commit: Day 26: Complete failure-analysis exercise

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 27 — Advanced Manual Testing

Objective: Perform a high-risk advanced test pass.

Study Topics

• long-duration

• low battery

• interference observations

• multiple devices

• transition testing

Hands-on Activities

1. Run long playback.

2. Test low-battery behavior if safe/practical.

3. Test repeated transitions.

4. Compare behavior across Android and Windows where applicable.

GitHub Deliverables

• 06-Test-Execution/Advanced-Test-Execution.xlsx

Suggested Commit: Day 27: Execute advanced manual testing

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 28 — Full Regression Cycle

Objective: Run a release-style regression cycle.

Study Topics

• smoke

• functional regression

• defect regression

• metrics

Hands-on Activities

1. Run the prioritized regression suite.

2. Calculate executed/passed/failed/blocked.

3. Summarize remaining risks and open defects.

GitHub Deliverables

• 06-Test-Execution/Regression-Cycle.xlsx

• 06-Test-Execution/Regression-Report.md

Suggested Commit: Day 28: Complete full regression cycle

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 29 — Test Summary Report

Objective: Prepare a release-style QA summary.

Study Topics

• executive summary

• coverage

• metrics

• defect summary

• risks

• recommendation

Hands-on Activities

1. Compile final metrics.

2. Summarize major findings.

3. State limitations and remaining risks.

4. Give a clear testing conclusion.

GitHub Deliverables

• 12-Test-Summary/Test-Summary-Report.md

Suggested Commit: Day 29: Prepare final Test Summary Report

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

Day 30 — Final QA Portfolio & Project Review

Objective: Complete the full 30-day portfolio and demonstrate the complete QA lifecycle.

Study Topics

• portfolio review

• lessons learned

• final quality assessment

Hands-on Activities

1. Review all folders.

2. Ensure links and filenames are clean.

3. Update root README.

4. Create final project summary.

5. Record lessons learned and future automation scope.

6. Tag or release the final version if desired.

GitHub Deliverables

• README.md

• 12-Test-Summary/Final-Project-Review.md

Suggested Commit: Day 30: Complete final QA portfolio

Completion Checklist

• Study completed

• Hands-on activity completed

• Actual observations recorded

• GitHub documentation updated

• Work reviewed for accuracy and clarity

• Commit created

5. Standard Test Case Template

Use this structure in Excel/Sheets:

Field	Purpose
Test Case ID	Unique identifier, e.g. TC-TWS-BT-001
Module	Bluetooth / Audio / Call / Connectivity
Requirement ID	Traceability reference
Title	Short, testable statement
Preconditions	Required setup before execution
Test Data	Audio file, device state, account/state if applicable
Steps	Numbered, reproducible actions
Expected Result	Observable expected behavior
Actual Result	Observed behavior during execution
Status	Pass / Fail / Blocked / Not Run
Severity	For failures/defects
Priority	Release/business urgency
Evidence	Screenshot/log/audio/evidence reference
Remarks	Additional observations

6. Standard Bug Report Template

- Bug ID and concise summary
- Environment: host, OS/version, DUT, connection state
- Preconditions
- Steps to reproduce
- Expected result
- Actual result
- Reproducibility/frequency
- Severity and priority
- Evidence
- Workaround if known
- Retest result
- Regression result

7. Test Execution Metrics

- Total test cases
- Executed test cases
- Passed, failed and blocked cases
- Pass percentage = Passed / Executed × 100
- Fail percentage = Failed / Executed × 100

• Total defects and defects by severity

• Open vs closed defects

• Retest results

• Regression results

8. Final Portfolio Checklist

• Research documentation complete

• Requirements documented

• Test Strategy complete

• Test Plan complete

• 100+ test cases targeted

• Execution records complete

• Defect reports documented

• Bug validation and regression documented

• RTM complete

• Python code review completed

• Failure analysis completed

• Test Summary Report completed

• Root README updated with final metrics and lessons learned

9. Recommended Next Stage After Day 30

Once the manual-testing portfolio is complete, extend the project into automation and deeper device testing: Python test utilities, Pytest, BLE discovery/GATT exploration, log parsing, ESP32-based Bluetooth experiments, automated report generation, and CI/CD. Automation should build on the manual test design rather than replace the foundational QA process.

10. Professional QA Mindset

Ask these questions for every feature:

• What is the requirement?

• What is the expected behavior?

• What did I actually observe?

• Can I reproduce the behavior?

• What evidence proves it?

• What is the impact?

• How should I retest it?

• What surrounding functionality should be included in regression?

Final principle: **Do not test only for success. Test for recovery, failure, repeatability and evidence.**

This manual is an independent educational training resource for TWS/Bluetooth audio QA skill development. It does not represent confidential, proprietary, or official procedures of any device manufacturer.